

1. List the inference rules used in propositional logic? Explain them in detail with suitable example. [9]

Introduction

Inference rules are the basic rules used in **Propositional Logic** to derive new conclusions from known facts. These rules help AI systems think logically and make correct decisions.

Inference rules are widely used in:

- Artificial Intelligence
- Expert systems
- Automated reasoning
- Theorem proving

Using inference rules, a computer can solve problems in a logical way just like humans.

Common Inference Rules in Propositional Logic

1. Modus Ponens (MP)

This is the most important inference rule.

It states that:

- If a condition is true
- And the implication is true
- Then the conclusion must also be true.

Rule

$P \rightarrow Q; P \vdash Q$

Explanation

If P implies Q and P is true, then Q is definitely true.

Example

- If it rains, roads become wet.
- It is raining.
- Therefore, roads are wet.

Importance

Used in many AI reasoning systems and expert systems.

2. Modus Tollens (MT)

This rule works in reverse direction.

It states that:

- If P implies Q
- And Q is false
- Then P must also be false.

Rule

$P \rightarrow Q; \neg Q \vdash \neg P$

Explanation

If the result does not happen, then the original condition was also not true.

Example

- If there is fire, there is smoke.
- There is no smoke.
- Therefore, there is no fire.

Importance

Useful in diagnosis and fault detection systems.

3. AND Rule (Conjunction)

This rule combines two true statements into one statement using AND.

Rule

$P; Q \vdash P \wedge Q$

Explanation

If both statements are true separately, then their combination is also true.

Example

- Ram is intelligent.
- Ram is hardworking.
- Therefore, Ram is intelligent and hardworking.

Importance

Used to combine facts in knowledge representation.

4. OR Rule (Addition)

This rule adds another statement using OR.

Rule

$P \vdash P \vee Q$

Explanation

If one statement is true, then the OR statement is also true.

Example

- Today is Sunday.
- Therefore, today is Sunday or Monday.

Importance

Used in logical expansion and reasoning.

5. Resolution Rule

Resolution is a powerful inference rule used in AI and theorem proving.

It combines two clauses to generate a new clause.

Rule

$P \vee Q, \neg P \vee R \vdash Q \vee R$

Explanation

The opposite literals cancel each other and remaining literals are combined.

Example

- A or B
- Not A or C
- Therefore, B or C

Importance

Widely used in automated theorem proving and Prolog.

6. Double Negation Rule

This rule says that removing two NOT operators gives the original statement.

Rule

$P \vdash \neg(\neg P)$

Example

- Not(not hot) = hot

Importance

Simplifies logical expressions.

Advantages of Inference Rules

- Helps in logical reasoning
- Used for decision making
- Helps AI systems derive conclusions
- Improves problem solving

Conclusion

Inference rules are the foundation of propositional logic. They help AI systems derive new knowledge from existing facts and make intelligent logical decisions.

2. Explain syntax and semantics of First Order Logic in detail. [8]

Introduction to First Order Logic (FOL)

First Order Logic is an advanced form of propositional logic used to represent real-world knowledge.

Unlike propositional logic, FOL can represent:

- Objects
- Relations
- Properties
- Quantities

It is widely used in:

- Artificial Intelligence
 - Expert systems
 - Robotics
 - Natural language processing
-

Syntax of First Order Logic

Syntax defines the rules for writing valid logical statements.

It tells:

- Which symbols are used
 - How formulas are formed
-

Components of Syntax

1. Constants

Constants represent specific objects or names.

Examples

- Ram
- India
- Carl

Explanation

A constant always refers to the same object.

2. Variables

Variables represent general or unknown objects.

Examples

- x
- y
- z

Explanation

Variables can take different values from the domain.

3. Predicates

Predicates represent properties or relationships between objects.

Examples

- Student(x)
- Loves(Ram, Mango)

- Greater(x,y)

Explanation

Predicates return either TRUE or FALSE.

4. Functions

Functions return a value or object.

Examples

- Father(Ram)
- Age(x)

Explanation

Functions map one object to another object.

5. Logical Connectives

These connect logical statements.

Symbol Meaning

$\wedge$	AND
$\vee$	OR
$\neg$	NOT
$\rightarrow$	IMPLIES
$\leftrightarrow$	IF AND ONLY IF

6. Quantifiers

Quantifiers specify quantity.

Universal Quantifier ($\forall$)

Means “for all”.

Example

$\forall x; \text{Human}(x) \rightarrow \text{Mortal}(x)$

Meaning

All humans are mortal.

Existential Quantifier ($\exists$)

Means “there exists”.

Example

$\exists x; \text{Student}(x)$

Meaning

There exists at least one student.

Semantics of First Order Logic

Semantics gives meaning to FOL statements.

It explains:

- Whether statements are true or false
 - How symbols relate to real-world objects
-

Important Concepts in Semantics

1. Domain

The set of all objects being discussed.

Example

- Students in a classroom
 - Animals in a zoo
-

2. Interpretation

Interpretation assigns meaning to constants, predicates, and functions.

Example

- $\text{Ram} \rightarrow \text{specific person}$
 - $\text{Student}(\text{Ram}) \rightarrow \text{TRUE}$
-

3. Truth Value

Every logical statement becomes either:

- TRUE
- FALSE

Example

- $\text{Human}(\text{Ram}) = \text{TRUE}$
-

Example of FOL Statement

$\forall x; \text{Bird}(x) \rightarrow \text{Fly}(x)$

Meaning

Every bird can fly.

If Tweety is a bird, then Tweety can fly.

Advantages of First Order Logic

- Represents complex knowledge
 - More expressive than propositional logic
 - Supports intelligent reasoning
 - Used in AI applications
-

Limitations of First Order Logic

- Complex for large systems
 - Requires more computation
 - Difficult to implement sometimes
-

Conclusion

Syntax defines the structure of FOL statements, while semantics provides their meaning. Together they help represent and reason about real-world knowledge in Artificial Intelligence.

3. Detail the algorithm for deciding entailment in propositional logic. [8]

Introduction

Entailment means checking whether a conclusion logically follows from a knowledge base.

If every model that satisfies the knowledge base also satisfies the query, then entailment is true.

It is written as:

$\text{KB} \models \alpha$

Where:

- KB = Knowledge Base
 - α = Query sentence
-

Purpose of Entailment Algorithm

The algorithm is used to:

- Verify logical conclusions

- Perform reasoning in AI
- Check correctness of statements

It is widely used in:

- Expert systems
- Automated reasoning
- Knowledge-based agents

Truth Table Algorithm for Entailment

The most common method for deciding entailment is the **Truth Table Method**.

This method checks all possible truth values.

Steps of the Algorithm

Step 1: Identify all symbols

List all propositional symbols in KB and query.

Example

- P
- Q

Step 2: Generate truth table

Create all possible combinations of TRUE and FALSE values.

P Q

T T

T F

F T

F F

Step 3: Evaluate Knowledge Base

Check which rows satisfy the knowledge base.

Step 4: Evaluate Query

Check whether query is true in all rows where KB is true.

Step 5: Decide Entailment

- If query is true in every model where KB is true $\rightarrow$ entailment holds.
- Otherwise $\rightarrow$ entailment fails.

Example

Knowledge Base

1. If P then Q
2. P is true

Query

Q

Logical Form

$P \rightarrow Q; P \models Q$

Truth Table

P Q $P \rightarrow Q$ KB Query(Q)

T T T T T

T F F F F

F T T F T

F F T F F

Explanation

The KB is true only in the first row.

In that row, Q is also true.

Therefore, KB entails Q.

Pseudocode

TT-ENTAILS(KB, α)

1. Generate all possible truth assignments
2. For each assignment:
 - If KB is true:
 - Check α
3. If α is true in all models where KB is true:
 - Return TRUE
- Else:
 - Return FALSE

Advantages

- Simple and easy to understand
- Gives correct result
- Complete method

Disadvantages

- Slow for many variables
- Truth table size grows rapidly

Applications

- AI reasoning
- Knowledge representation
- Logic programming
- Expert systems

4. Explain knowledge representation structure and compare them. [9]

Introduction

Knowledge Representation (KR) is a method used in Artificial Intelligence to store knowledge in a form that a computer can understand and use for reasoning.

It helps machines:

- Store facts
- Solve problems
- Make decisions
- Understand relationships

A good knowledge representation technique should be:

- Easy to understand
- Efficient
- Flexible
- Capable of reasoning

Types of Knowledge Representation Structures

1. Logical Representation

Knowledge is represented using logical statements and rules.

It uses:

- Propositional Logic
- First Order Logic

Example

“All humans are mortal.”

\forall x; Human(x) \rightarrow Mortal(x)

Advantages

- Clear and precise
- Supports reasoning

Disadvantages

- Difficult for large systems
- Complex syntax

2. Semantic Network

Knowledge is represented as a graph containing:

- Nodes $\rightarrow$ objects
- Links $\rightarrow$ relationships

Example

Bird $\rightarrow$ can fly

Advantages

- Easy visualization
- Represents relationships clearly

Disadvantages

- Difficult for complex reasoning

3. Frame Representation

Frames are data structures used to represent stereotyped situations.

A frame contains:

- Slots
- Values

Example

Frame Slots

Student Name, Roll No, Marks

Advantages

- Easy organization
- Supports inheritance

Disadvantages

- Less flexible for dynamic knowledge

4. Production Rules

Knowledge is represented using IF–THEN rules.

Example

IF fever THEN disease = flu

Advantages

- Simple and easy
- Widely used in expert systems

Disadvantages

- Large number of rules may become difficult to manage

Comparison of Knowledge Representation Structures

Technique	Representation Style	Advantages	Disadvantages
Logical Representation	Facts and formulas	Accurate reasoning	Complex
Semantic Network	Graph structure	Easy visualization	Weak reasoning
Frame Representation	Slots and values	Organized structure	Less flexible
Production Rules	IF–THEN rules	Easy to understand	Rule explosion

Applications of Knowledge Representation

- Expert systems
 - Robotics
 - Medical diagnosis
 - Natural language processing
 - Intelligent agents
-

5. i. What is an Agent? Name any 5 agents around you. Explain Knowledge-based agent with Wumpus World. [9]

What is an Agent?

An Agent is anything that:

- Perceives the environment using sensors
- Acts upon the environment using actuators

An intelligent agent takes decisions to achieve goals.

Examples of Agents Around Us

1. Mobile phone assistant
2. Robot vacuum cleaner
3. ATM machine
4. Self-driving car
5. Air conditioner thermostat

Knowledge-Based Agent

A Knowledge-Based Agent is an intelligent agent that uses stored knowledge and logical reasoning to take decisions.

It contains:

- Knowledge Base (KB)
- Inference Engine

Components of Knowledge-Based Agent

1. Knowledge Base

Stores:

- Facts
- Rules
- Information about environment

2. Inference Engine

Applies logical reasoning on stored facts to derive conclusions.

Working of Knowledge-Based Agent

1. Receives information from environment
2. Stores facts in KB
3. Applies reasoning
4. Decides suitable action

Wumpus World

Wumpus World is a famous AI problem used to test logical reasoning.

It is a cave represented by a grid.

The world contains:

- Wumpus (monster)
- Pits
- Gold
- Agent

Goal of the Agent

The agent must:

- Find the gold
- Avoid pits
- Avoid Wumpus
- Safely return back

Rules in Wumpus World

- Breeze near pit
- Stench near Wumpus
- Glitter near gold

Example

If there is breeze in a room, the agent infers that a pit may be nearby.

Using logical reasoning, the agent decides safe movement.

Advantages of Knowledge-Based Agent

- Intelligent reasoning
 - Better decision making
 - Learns from knowledge
 - Solves complex problems
-

5. ii. List and explain in short the various steps of knowledge engineering process. [9]

Introduction

Knowledge Engineering is the process of building knowledge-based systems.

It involves:

- Collecting knowledge
- Representing knowledge
- Applying reasoning

The main aim is to develop intelligent systems.

Steps in Knowledge Engineering Process

1. Identify the Problem

The first step is to define:

- Problem domain
- Goals
- Requirements

Example

Medical diagnosis system.

2. Acquire Knowledge

Knowledge is collected from:

- Experts
- Books
- Databases
- Internet

Example

Collecting disease symptoms from doctors.

3. Represent Knowledge

Knowledge is stored using:

- Rules
- Logic
- Frames
- Semantic networks

Example

IF fever THEN flu.

4. Implement the System - The knowledge base and inference engine are implemented in software.

5. Perform Reasoning - The system applies inference rules to derive conclusions.

6. Test and Validate

The system is checked for:

- Accuracy
- Correctness
- Performance

7. Update Knowledge - Knowledge is updated regularly as new information becomes available.

Diagram of Knowledge Engineering Process

Problem Identification → Knowledge Acquisition → Knowledge Representation → Implementation → Reasoning → Testing → Updating

Advantages

- Improves decision making
- Stores expert knowledge
- Reduces human effort

Applications

- Expert systems
 - Robotics
 - AI assistants
 - Medical diagnosis
-

6. Consider the following axioms: [9]

Statements

1. If a triangle is isosceles, then its two sides AB and AC are equal.
2. If AB and AC are equal, then angle B and angle C are equal.
3. ABC is an equilateral triangle.

Represent these facts in predicate logic. Explain inference in propositional logic.

Predicate Logic Representation

Statement 1

“If a triangle is isosceles, then AB and AC are equal.”

$\text{Isosceles}(ABC) \rightarrow \text{Equal}(AB, AC)$

Statement 2

“If AB and AC are equal, then angle B and angle C are equal.”

$\text{Equal}(AB, AC) \rightarrow \text{Equal}(\angle B, \angle C)$

Statement 3

“ABC is an equilateral triangle.”

$\text{Equilateral}(ABC)$

Additional Rule

An equilateral triangle is also isosceles.

$\text{Equilateral}(ABC) \rightarrow \text{Isosceles}(ABC)$

Final Inference

From the above facts:

- ABC is equilateral
- Therefore ABC is isosceles
- Hence $AB = AC$
- Therefore $\angle B = \angle C$

Inference in Propositional Logic

Inference means deriving new conclusions from known facts using logical rules.

AI systems use inference to perform reasoning and decision making.

Common Inference Rules

1. Modus Ponens

If:

- $(P \rightarrow Q)$
- (P)

Then:

- (Q)

Example

- If it rains, roads become wet.
- It is raining.
- Therefore roads are wet.

2. Modus Tollens

If:

- $(P \rightarrow Q)$
- Not Q

Then:

- Not P

Example

- If there is fire, there is smoke.
- No smoke.
- Therefore no fire.

3. Resolution

Combines clauses to derive new clause.

Example

- A or B
- Not A or C
- Therefore B or C

Importance of Inference

- Helps logical reasoning
 - Used in expert systems
 - Supports intelligent decision making
-

7. Write the following sentences in First Order Logic (FOL) using quantifiers. [9]

Introduction

First Order Logic (FOL) is used in Artificial Intelligence to represent facts, objects, properties, and relationships logically.

Quantifiers are special symbols used in FOL.

Types of Quantifiers

1. Universal Quantifier ($\forall$)

Means “for all”.

2. Existential Quantifier ($\exists$)

Means “there exists”.

i) Every number is either negative or has a square root.

Explanation

This statement says that for every number:

- Either the number is negative
- Or it has a square root

Predicate Logic Representation

$\forall x; [\text{Negative}(x) \vee \text{HasSquareRoot}(x)]$

Meaning

For every number x:

- x is negative
- OR
- x has a square root.

ii) Every connected and circuit-free graph is a tree.

Explanation

A graph becomes a tree if:

- It is connected
- It has no cycles or circuits

Predicate Logic Representation

$\forall x; [(\text{Connected}(x) \wedge \text{CircuitFree}(x)) \rightarrow \text{Tree}(x)]$

Meaning

For every graph x:

- If x is connected
- AND

- x is circuit-free

Then:

- x is a tree.

iii) Some people are either religious or pious.

Explanation

This statement says that there exists at least one person who is:

- Religious
- OR
- Pious

Predicate Logic Representation

$\exists x; [\text{Religious}(x) \vee \text{Pious}(x)]$

Meaning

There exists a person x such that:

- x is religious
- OR
- x is pious.

iv) There is a barber who shaves all men in the town who do not shave themselves.

Explanation

This is a famous logical statement called the Barber Problem.

It states:

- There exists a barber
- The barber shaves all men
- Who do not shave themselves

Predicate Logic Representation

$\exists x; [\text{Barber}(x) \wedge \forall y; ((\text{Man}(y) \wedge \neg \text{Shaves}(y,y)) \rightarrow \text{Shaves}(x,y))]$

Meaning

There exists a barber x such that:

- x is a barber
- AND
- For every man y,
if y does not shave himself,
then x shaves y.

8. What is resolution? Solve the following statement using resolution algorithm. Draw suitable resolution graph. [9]

Introduction to Resolution

Resolution is an important inference rule used in Artificial Intelligence and automated theorem proving.

It is used to:

- Prove statements logically
- Derive conclusions
- Perform reasoning automatically

Resolution works by:

1. Converting statements into clause form
2. Applying resolution rule
3. Deriving contradiction or conclusion

Resolution Rule

$P \vee Q, \neg P \vee R \vdash Q \vee R$

Given Statements

1. Rajesh likes all kinds of food.
2. Apple and vegetables are food.

3. Anything anyone eats and is not killed is food.

4. Ajay eats peanuts and is still alive.

Goal: Prove that Rajesh likes bananas.

Step 1: Convert Statements into Predicate Logic

Statement 1

“Rajesh likes all kinds of food.”

$\forall x; \text{Food}(x) \rightarrow \text{Likes}(\text{Rajesh}, x)$

Statement 2

“Apple and vegetables are food.”

$\text{Food}(\text{Apple}) \wedge \text{Food}(\text{Vegetable})$

Statement 3

“Anything anyone eats and is not killed is food.”

$\forall x \forall y; [(\text{Eats}(y, x) \wedge \text{Alive}(y)) \rightarrow \text{Food}(x)]$

Statement 4

“Ajay eats bananas and is alive.”

$\text{Eats}(\text{Ajay}, \text{Banana}) \wedge \text{Alive}(\text{Ajay})$

Goal

Prove:

$\text{Likes}(\text{Rajesh}, \text{Banana})$

Step 2: Convert into Clause Form

Clause 1

From statement 1:

$\neg \text{Food}(x) \vee \text{Likes}(\text{Rajesh}, x)$

Clause 2

From statement 3:

$\neg \text{Eats}(y, x) \vee \neg \text{Alive}(y) \vee \text{Food}(x)$

Clause 3

$\text{Eats}(\text{Ajay}, \text{Banana})$

Clause 4

$\text{Alive}(\text{Ajay})$

Step 3: Negate Goal

Negated Goal:

$\neg \text{Likes}(\text{Rajesh}, \text{Banana})$

Step 4: Apply Resolution

Resolve Clause 2 with Clause 3

From:

- $\text{Eats}(\text{Ajay}, \text{Banana})$
- $\neg \text{Eats}(y, x) \vee \neg \text{Alive}(y) \vee \text{Food}(x)$

We get:

$\neg \text{Alive}(\text{Ajay}) \vee \text{Food}(\text{Banana})$

Resolve with Clause 4

From:

- $\text{Alive}(\text{Ajay})$
- $\neg \text{Alive}(\text{Ajay}) \vee \text{Food}(\text{Banana})$

We get:

$\text{Food}(\text{Banana})$

Resolve with Clause 1

From:

- $\text{Food}(\text{Banana})$
- $\neg \text{Food}(x) \vee \text{Likes}(\text{Rajesh}, x)$

We get:

Likes(Rajesh,Banana)

Resolve with Negated Goal

From:

- Likes(Rajesh,Banana)
- $\neg$ Likes(Rajesh,Banana)

We get contradiction:

\Box

Resolution Graph

Eats(Ajay,Banana) Alive(Ajay)

 \ /

Food(Banana)

 | |

Likes(Rajesh,Banana)

 |

$\neg$ Likes(Rajesh,Banana)

 |

Contradiction

9. Consider the following axioms: [9]

1. If a triangle is equilateral then it is isosceles.
2. If a triangle is isosceles, then its two sides AB and AC are equal.
3. If AB and AC are equal, then angle B and angle C are equal.
4. ABC is an equilateral triangle.

Represent these facts in predicate logic.

Introduction

Predicate Logic is used in AI to represent mathematical and logical relationships.

It uses:

- Predicates
- Variables
- Quantifiers
- Logical connectives

Predicate logic provides clear and precise representation of knowledge.

Step 1: Define Predicates

Predicate Meaning

Equilateral(x) x is equilateral

Isosceles(x) x is isosceles

Equal(x,y) x and y are equal

Step 2: Represent Statements in Predicate Logic

Statement 1

“If a triangle is equilateral then it is isosceles.”

$\forall x; [\text{Equilateral}(x) \rightarrow \text{Isosceles}(x)]$

Statement 2

“If a triangle is isosceles, then sides AB and AC are equal.”

$\text{Isosceles}(ABC) \rightarrow \text{Equal}(AB,AC)$

Statement 3

“If AB and AC are equal, then angle B and angle C are equal.”

$\text{Equal}(AB,AC) \rightarrow \text{Equal}(\angle B, \angle C)$

Statement 4

“ABC is an equilateral triangle.”

Equilateral(ABC)

Inference from the Statements

From:

- ABC is equilateral
- Equilateral triangles are isosceles

Therefore:

Isosceles(ABC)

Then:

Equal(AB,AC)

Finally:

Equal($\angle B, \angle C$)

10. What is Knowledge Representation using propositional logic? Compare propositional and predicate logic. [9]

Introduction

Knowledge Representation (KR) is a technique used in Artificial Intelligence to store information in a form that a computer can understand and use.

It helps AI systems:

- Store facts
- Solve problems
- Perform reasoning
- Make decisions

One important method of knowledge representation is **Propositional Logic**.

Knowledge Representation using Propositional Logic

Propositional Logic represents knowledge using propositions or statements.

A proposition is a statement that is either:

- TRUE
- FALSE

It uses logical symbols to combine statements.

Basic Logical Symbols

Symbol Meaning

$\wedge$ AND

$\vee$ OR

$\neg$ NOT

$\rightarrow$ IMPLIES

$\leftrightarrow$ IF AND ONLY IF

Example of Propositional Logic

Let:

- P = “It is raining”
- Q = “Roads are wet”

Then:

$P \rightarrow Q$

Meaning:

“If it rains, roads become wet.”

Features of Propositional Logic

- Uses simple statements
- Statements are true or false
- Supports logical reasoning

- Easy to understand

Advantages of Propositional Logic

1. Simple and easy representation
2. Easy implementation
3. Useful for basic reasoning
4. Used in automated reasoning systems

Limitations of Propositional Logic

1. Cannot represent relations between objects
2. Cannot use variables or quantifiers
3. Difficult for complex real-world problems
4. Less expressive than predicate logic

Predicate Logic

Predicate Logic is an advanced form of propositional logic.

It can represent:

- Objects
- Relations
- Properties
- Quantifiers

Example of Predicate Logic

$\forall x; \text{Human}(x) \rightarrow \text{Mortal}(x)$

Meaning:

“All humans are mortal.”

Comparison between Propositional Logic and Predicate Logic

Propositional Logic	Predicate Logic
Represents simple facts	Represents complex facts
No variables	Uses variables
No quantifiers	Uses quantifiers
Less expressive	More expressive
Easy to implement	More complex
Cannot represent relations	Can represent relations
Example: $P \rightarrow Q$	Example: $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Applications

Propositional Logic

- Simple AI systems
- Circuit design
- Basic reasoning systems

Predicate Logic

- Expert systems
- Robotics
- Natural language processing
- Knowledge-based systems

11. Write the following sentences in FOL using quantifiers. [9]

Introduction

First Order Logic (FOL) is used to represent knowledge mathematically using:

- Predicates
- Variables

- Quantifiers

Quantifiers:

- Universal Quantifier ($\forall$) $\rightarrow$ “for all”
- Existential Quantifier ($\exists$) $\rightarrow$ “there exists”

i) All birds fly

Predicate Logic Representation

$\forall x; [\text{Bird}(x) \rightarrow \text{Fly}(x)]$

Meaning

For every object x:

- If x is a bird
- Then x can fly

ii) Some boys play cricket

Predicate Logic Representation

$\exists x; [\text{Boy}(x) \wedge \text{PlaysCricket}(x)]$

Meaning

There exists at least one boy who plays cricket.

iii) A first cousin is a child of a parent's sibling

Predicate Logic Representation

$\forall x \forall y; [\text{FirstCousin}(x,y) \rightarrow \text{Child}(x, \text{Sibling}(z))]$

Meaning

If x and y are first cousins, then x is the child of a sibling of y's parent.

iv) You can fool all the people some of the time, and some of the people all the time, but you cannot fool all the people all the time.

Predicate Logic Representation

First Part

“You can fool all the people some of the time.”

$\forall x; \exists t; \text{Fool}(x,t)$

Meaning:

For every person x, there exists some time t when x can be fooled.

Second Part

“You can fool some of the people all the time.”

$\exists x; \forall t; \text{Fool}(x,t)$

Meaning:

There exists some person x who can be fooled all the time.

Third Part

“You cannot fool all the people all the time.”

$\neg \forall x \forall t; \text{Fool}(x,t)$

Meaning:

It is not possible to fool everyone at all times.

12. Explain Wumpus world environment giving its PEAS description. [9]

Introduction

Wumpus World is a famous problem in Artificial Intelligence used to test logical reasoning and intelligent decision making.

It is a grid-based environment containing:

- Wumpus (monster)
- Pits
- Gold
- Agent

The agent must find the gold and safely come out without falling into pits or being killed by the Wumpus.

Features of Wumpus World

- The world consists of rooms arranged in a grid.
- Some rooms contain pits.
- One room contains the Wumpus.
- One room contains gold.
- The agent starts from the first room.

Rules of Wumpus World

Situation	Indication
Pit nearby	Breeze
Wumpus nearby	Stench
Gold nearby	Glitter

Goal of the Agent

The intelligent agent must:

1. Find the gold
2. Avoid pits
3. Avoid Wumpus
4. Return safely

Working of Wumpus Agent

The agent:

- Observes environment
- Uses logical reasoning
- Decides safe movement
- Updates knowledge base

PEAS Description of Wumpus World

PEAS stands for:

- Performance Measure
- Environment
- Actuators
- Sensors

1. Performance Measure

Measures success of the agent.

Goals

- Find gold
- Avoid death
- Minimize moves
- Exit safely

Performance Criteria

- +1000 for getting gold
- -1000 for death
- Small penalty for each move

2. Environment

The environment includes:

- Grid rooms
- Pits
- Wumpus
- Gold
- Walls

Nature of Environment

- Partially observable

- Sequential
- Static
- Discrete

3. Actuators

Actuators are actions performed by the agent.

Actions

- Move forward
- Turn left
- Turn right
- Grab gold
- Shoot arrow
- Climb out

4. Sensors

Sensors help the agent perceive the environment.

Sensors in Wumpus World

Sensor	Meaning
Breeze	Pit nearby
Stench	Wumpus nearby
Glitter	Gold nearby
Bump	Hit wall
Scream	Wumpus killed

Example

If the agent senses:

- Breeze → there may be a pit nearby
- Stench → Wumpus may be nearby
- Glitter → gold is present

The agent uses reasoning to move safely.

Advantages of Wumpus World

- Tests logical reasoning
- Demonstrates knowledge-based agents
- Helps understand AI decision making

Applications

- Robotics
- Intelligent systems
- Game AI
- Knowledge representation research

13. Explain different inference rules in FOL with suitable example. [8]

Introduction

Inference rules in First Order Logic (FOL) are used to derive new conclusions from known facts.

They help AI systems:

- Perform reasoning
- Solve problems
- Make decisions logically

Inference rules are important in:

- Expert systems
- Automated theorem proving
- Knowledge-based systems

Different Inference Rules in FOL

1. Universal Instantiation (UI)

This rule states that:

If something is true for all objects, then it is true for a particular object.

General Form

$\forall x; P(x) \vdash P(a)$

Example

“All humans are mortal.”

$\forall x; \text{Human}(x) \rightarrow \text{Mortal}(x)$

If Ram is human:

$\text{Human}(\text{Ram})$

Then:

$\text{Mortal}(\text{Ram})$

Explanation

Since every human is mortal and Ram is human, Ram must be mortal.

2. Existential Instantiation (EI)

This rule states that:

If there exists an object having a property, then we can assign a name to that object.

General Form

$\exists x; P(x) \vdash P(a)$

Example

“There exists a student who is intelligent.”

$\exists x; [\text{Student}(x) \wedge \text{Intelligent}(x)]$

Assume the student is Ram:

$\text{Student}(\text{Ram}) \wedge \text{Intelligent}(\text{Ram})$

Explanation

Since at least one intelligent student exists, we can consider Ram as that student.

3. Modus Ponens

If:

- $P \text{ implies } Q$
- $P \text{ is true}$

Then:

- $Q \text{ is true}$

General Form

$P \rightarrow Q, P \vdash Q$

Example

- If it rains, roads become wet.
- It is raining.

Therefore:

- Roads are wet.

Importance

Most commonly used inference rule.

4. Modus Tollens

If:

- $P \text{ implies } Q$
- $Q \text{ is false}$

Then:

- $P \text{ is false}$

General Form

$P \rightarrow Q, \neg Q \vdash \neg P$

Example

- If there is fire, there is smoke.
- There is no smoke.

Therefore:

- There is no fire.

5. Resolution Rule

Resolution combines clauses to derive new conclusions.

General Form

$P \vee Q, \neg P \vee R \vdash Q \vee R$

Example

- A or B
- Not A or C

Therefore:

- B or C

Advantages of Inference Rules

- Helps logical reasoning
- Supports intelligent systems
- Used in theorem proving
- Helps derive new knowledge

Applications

- Expert systems
- Robotics
- AI reasoning systems
- Medical diagnosis systems

14. Write propositional logic for the statements. [10]

Introduction

Propositional Logic represents statements using logical symbols and propositions.

A proposition is a statement that is either:

- TRUE
- FALSE

Logical connectives are used to combine propositions.

i) “All birds fly”

Predicate Logic Representation

Since “all” involves quantification, it is represented better using predicate logic.

$\forall x; [Bird(x) \rightarrow Fly(x)]$

Explanation

For every object x:

- If x is a bird
- Then x can fly

Example

If Sparrow is a bird, then Sparrow flies.

ii) “Every man respects his parents”

Predicate Logic Representation

$\forall x \forall y; [(Man(x) \wedge Parent(y,x)) \rightarrow Respect(x,y)]$

Explanation

For every person x and y:

- If x is a man
- And y is parent of x
- Then x respects y

Example

If Ram is a man and Shyam is Ram’s father, then Ram respects Shyam.

Importance of Predicate Logic

- Represents relations clearly

- Supports quantifiers
- More expressive than propositional logic

15. Differentiate between propositional logic and First Order Logic. [7]

Introduction

Both propositional logic and First Order Logic (FOL) are used in Artificial Intelligence for knowledge representation and reasoning.

However, FOL is more powerful and expressive than propositional logic.

Difference between Propositional Logic and First Order Logic

Propositional Logic	First Order Logic
Represents simple statements	Represents objects and relations
Uses propositions only	Uses predicates and variables
No quantifiers	Uses quantifiers
Less expressive	More expressive
Cannot represent relations	Can represent relations
Easier to implement	More complex
Limited reasoning ability	Strong reasoning ability
Statements are either true or false	Statements include objects and properties
Example: $P \rightarrow Q$	Example: $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Example of Propositional Logic

Let:

- P = "It is raining"
- Q = "Roads are wet"

Then:

$P \rightarrow Q$

Example of First Order Logic

$\forall x; \text{Human}(x) \rightarrow \text{Mortal}(x)$

Meaning:

All humans are mortal.

Advantages of Propositional Logic

- Simple
- Easy implementation
- Fast reasoning

Advantages of First Order Logic

- More expressive
- Represents real-world knowledge
- Better reasoning capability

Applications

Propositional Logic

- Digital circuits
- Simple expert systems
- Logical reasoning

First Order Logic

- AI systems
- Natural language processing
- Robotics
- Knowledge representation